Software Requirements Specification — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the Online Examination Portal. It is the single source for the GitHub backlog (one backlog item per requirement), the Use Case Diagram (`Documents/Use_Case_Diagram.pdf`), the Use Case Flow (`Documents/Use_Case_Flow.pdf`), the Software Architecture Document (`Documents/SAD.pdf`) and the Test Plan (`Documents/Test_Plan.pdf`).

1.2 Scope

Faculty author multiple-choice questions into a tagged question bank, assemble them into an examination and schedule it for a fixed window. Enrolled Students attempt the examination under a server-enforced timer with autosave and resume. Every submission is evaluated automatically against the stored answer key, and results are released when Faculty publishes them.

Out of scope: descriptive answers and manual grading · webcam or AI proctoring · biometric identity verification · payments · adaptive testing · native mobile applications · offline mode · plagiarism detection.

Why 30 functional requirements. The first requirements baseline (M1, 19 Aug) held 83 functional and 27 non-functional requirements — more than a team of three can build, test at 8–10 test cases per use case, and demonstrate in two one-week sprints. On 2 Sep it was reduced to the demonstration-critical core: every requirement needed to take one examination through its whole lifecycle (set up → attempt → evaluate → publish) was kept, and conveniences such as automatic paper generation, CSV import and export, multiple-correct questions and per-student shuffling were dropped. The four security requirements NFR-011 … NFR-014 were added on 15 Oct from the project's security analysis.

1.3 Actors

Actor	Description
Student	An enrolled candidate who sits examinations and views published results
Faculty	Authors questions and examinations, schedules them, publishes results
Administrator	Manages courses, enrolls students and reviews the audit trail

Actor	Description
Clock	The server clock — triggers automatic submission when an attempt's time runs out

1.4 How to read the requirement tables

The tables follow the course requirements-table template: **Req ID, Type, Description ("The system shall …"), Priority (High / Medium / Low), Acceptance Criteria (measurable pass/fail), Rationale and Comments**. For a functional requirement the Comments column names the use case that realises it, the sprint that implements it and how it is verified; for a non-functional requirement, the sprint. Non-functional requirements are classed as **Product**, **Organisational** or **External** in the Type column. Requirement IDs are sequential with no gaps: `FR-001` … `FR-030` and `NFR-001` … `NFR-014`.

2. Functional Requirements

2.1 User Management & Authentication

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall allow a visitor to register a Student account by submitting full name, institutional email, roll number and password.	High	Pass: a submission with all four valid fields creates a Student account; a missing field, a non-institutional email, or a duplicate email or roll number is rejected with an inline error and no account is created.	Students must be able to join without waiting for an administrator.	UC-01 · Sprint 1 · Verified by: Functional test
FR-002	Functional	The system shall authenticate a user against email and password and establish a session on success.	High	Pass: correct credentials open the dashboard for the user's role; a wrong password shows "Invalid email or password" and no session is created.	Every other function depends on a known, signed-in user.	UC-02 · Sprint 1 · Verified by: Functional test
FR-003	Functional	The system shall assign exactly one role — Student, Faculty or Administrator — to every account, and shall restrict every screen and endpoint to the roles authorised for it.	High	Pass: a Student requesting a Faculty URL directly receives HTTP 403; an unauthenticated request to any protected URL is redirected to sign-in (HTTP 302) or refused (HTTP 401).	Hiding a button is not access control; the server must refuse.	UC-02 · Sprint 1 · Verified by: Security audit / test

2.2 Course & Enrollment Management

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-004	Functional	The system shall allow an Administrator to create, edit and archive a Course identified by course code, title and semester, and to assign one or more Faculty accounts to it.	High	Pass: a Course created with code, title and semester appears in the course list with its assigned Faculty; a duplicate course code is rejected; an archived Course accepts no new examinations.	Examinations, questions and enrollments all hang off a Course.	UC-03 · Sprint 1 · Verified by: Functional test
FR-005	Functional	The system shall allow Faculty or an	High	Pass: after enrolling a Student, that Student	Enrollment decides who may	UC-04 · Sprint 1 ·

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
		Administrator to enroll Student accounts into a Course individually.		appears on the Course roster; enrolling the same Student twice leaves exactly one enrollment.	sit which examination.	Verified by: Functional test
FR-006	Functional	The system shall permit a Student to view and attempt only those examinations that belong to Courses in which the Student is enrolled.	High	Pass: a Student not enrolled in the Course does not see its examination and receives HTTP 403 on a direct URL to it.	Stops students sitting papers of courses they are not registered for.	UC-04 · Sprint 1 · Verified by: Integration test

2.3 Question Bank Management

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-007	Functional	The system shall allow Faculty to create a single-correct multiple-choice question consisting of a stem, between 2 and 6 options, and exactly one option marked correct.	High	Pass: a question with 4 options and one correct option is saved; 1 option, 7 options, zero correct or two correct options are each rejected with an error.	MCQ authoring is the core content of the system.	UC-05 · Sprint 1 · Verified by: Functional test
FR-008	Functional	The system shall require every question to carry a Course, a topic label, a difficulty of Easy, Medium or Hard, positive marks, and negative marks.	High	Pass: saving a question with any of the five fields missing, a difficulty outside the three values, or negative positive-marks is rejected.	Tags make the bank searchable; marks drive scoring.	UC-05 · Sprint 1 · Verified by: Data validation test
FR-009	Functional	The system shall prevent modification or deletion of a question that belongs to a published or attempted examination, and shall instead create a new version of that question.	Medium	Pass: editing a question used in a published examination creates version 2; the examination still shows version 1 unchanged; deleting it is refused.	Changing a question under a live paper would make past scores wrong.	UC-05 · Sprint 2 · Verified by: Database integrity test

2.4 Examination Authoring

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-010	Functional	The system shall allow Faculty to create an Examination with a title, Course, duration in minutes, declared total marks, instructions, and the policy values: negative marking (on/off and value), question and option shuffling, and pass mark.	High	Pass: an Examination saved with all fields is listed in Draft state with the entered values; a duration of 0 or a pass mark above 100% is rejected.	Each paper needs its own rules.	UC-06 · Sprint 1 · Verified by: Functional test
FR-011	Functional	The system shall allow Faculty to add questions to an Examination by selecting them from the question bank.	High	Pass: questions selected from the bank appear in the Examination in the selected order; a question from another Course cannot be added.	Papers are built by reuse, not retyping.	UC-06 · Sprint 1 · Verified by: Functional test
FR-012	Functional	The system shall hold an Examination in Draft state until Faculty publishes it, and shall make only published Examinations visible to Students.	High	Pass: a Draft examination is not visible on any Student dashboard; after Publish it is visible to enrolled Students.	Faculty must be able to prepare a paper privately.	UC-06 · Sprint 1 · Verified by: State transition test

2.5 Scheduling & Access Control

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-013	Functional	The system shall allow Faculty to schedule a published Examination by setting a window start date-time and end date-time.	High	Pass: the window is saved and shown on the Examination; an end time earlier than the start time is rejected.	The window decides when the paper can be taken.	UC-07 · Sprint 1 · Verified by: Functional test
FR-014	Functional	The system shall present to each Student a dashboard listing the Upcoming, Live and Completed Examinations of the Courses in which the Student is enrolled.	High	Pass: with server time before, inside and after the window, the examination appears under Upcoming, Live and Completed respectively.	Students need one place to find their papers.	UC-07 · Sprint 2 · Verified by: Functional test
FR-015	Functional	The system shall permit a Student to begin an attempt only when the current server time lies within the window, the Student is enrolled, and the Student has no previously submitted attempt for that Examination.	High	Pass: Start is refused with a message before the window opens, after it closes, for a non-enrolled Student and for a second attempt; it succeeds otherwise.	Enforces fairness: same window, one attempt each.	UC-08 · Sprint 1 · Verified by: System test

2.6 Examination Delivery (Attempt)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-016	Functional	The system shall create an attempt record stamped with the server start time when a Student begins an Examination, and shall deliver the paper only after that record exists.	High	Pass: the attempt row with a server start time exists before the first question is returned; no question is returned if the row cannot be created.	No one can see the paper without a timed attempt on record.	UC-08 · Sprint 1 · Verified by: Sequence test
FR-017	Functional	The system shall display a countdown timer whose remaining time is derived from the server clock and the attempt start time, and shall not rely on the client clock.	High	Pass: moving the client PC clock forward by 30 minutes does not change the remaining time shown after a page reload.	A client clock can be tampered with.	UC-08 · Sprint 1 · Verified by: Security / unit test
FR-018	Functional	The system shall display one question at a time with a navigation palette showing each question as Not Visited, Not Answered, Answered or Marked for Review.	High	Pass: answering, skipping and marking questions changes each palette cell to the matching state, and the states survive a page reload.	Lets students pace and review a long paper.	UC-08 · Sprint 2 · Verified by: UI / state test
FR-019	Functional	The system shall persist every response to the server within 5 seconds of the Student selecting it, without any explicit save action.	High	Pass: 5 seconds after selecting an option, the response is present in the database; killing the browser then loses no response older than 5 s.	A dropped connection must not cost a student their answers.	UC-09 · Sprint 2 · Verified by: Resilience / API test
FR-020	Functional	The system shall automatically submit an attempt, retaining all responses persisted up to that instant, when the remaining time reaches zero.	High	Pass: an attempt left open past its duration is in Submitted state with all saved responses, and no response saved after expiry is accepted.	Time limits must be enforced the same for every student.	UC-10 · Sprint 1 · Verified by: Automation test
FR-021	Functional	The system shall allow a Student whose session was interrupted to resume the same attempt, with remaining time computed from	High	Pass: closing the browser 10 minutes into a 60-minute attempt and reopening it 5 minutes later shows	Interruptions happen; they must neither erase work nor add time.	UC-09 · Sprint 2 · Verified by: Fault-injection test

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
		the original server start time and all saved responses restored.		about 45 minutes left and every earlier answer.		

2.7 Automatic Evaluation & Scoring

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-022	Functional	The system shall evaluate every submitted attempt automatically, without any human intervention.	High	Pass: every attempt in Submitted state, whether submitted manually or automatically, has a stored score without any Faculty action.	Removes manual grading entirely.	UC-11 · Sprint 1 · Verified by: Unit test
FR-023	Functional	The system shall award the question's full positive marks for a correct response, deduct the configured negative marks for an incorrect response, and award zero with no deduction for an unanswered question.	High	Pass: with +2 / −0.5 marking, 10 correct, 4 wrong and 6 unanswered scores exactly 18.0.	The scoring rule must be exact and predictable.	UC-11 · Sprint 1 · Verified by: Calculation test
FR-024	Functional	The system shall compute, for every evaluated attempt, the total score, the percentage and a Pass or Fail outcome against the Examination's pass mark.	High	Pass: 20 out of 50 with a 40% pass mark is stored as 40.0% and Pass; 19.5 out of 50 is 39.0% and Fail.	Students and Faculty need the outcome, not only the raw score.	UC-11 · Sprint 2 · Verified by: Calculation test

2.8 Results

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-025	Functional	The system shall withhold every evaluated result from the Student until Faculty publishes the results of that Examination.	High	Pass: before publication the Student sees only "Submitted" with no score, and the results URL returns HTTP 403.	Faculty decide when results are released.	UC-12 · Sprint 2 · Verified by: Access control test
FR-026	Functional	The system shall allow Faculty to publish and to unpublish the results of an Examination.	High	Pass: after Publish the scorecard is visible to the Student; after Unpublish it is hidden again.	Faculty need to correct a premature release.	UC-12 · Sprint 2 · Verified by: Functional test
FR-027	Functional	The system shall display to a Student, for a published result, the total score, the percentage, the Pass or Fail outcome and the marks obtained on each question.	High	Pass: every figure on the scorecard equals the stored evaluation exactly, including per-question marks.	Students should see how the score was reached.	UC-12 · Sprint 2 · Verified by: Functional test

2.9 Integrity & Audit

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-028	Functional	The system shall record, with a server-side timestamp, every occurrence of the examination window losing focus during an active attempt.	Medium	Pass: switching tabs three times during an attempt creates three focus-loss records with server timestamps, visible to Faculty.	Gives Faculty evidence; a human decides.	UC-13 · Sprint 2 · Verified by: Event capture test

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-029	Functional	The system shall exclude all correct-answer data from every response delivered to a Student during an active attempt.	High	Pass: the page source and every network response of an attempt contain no correct-option field or value.	Leaking the key in the page would void the exam.	UC-13 · Sprint 1 · Verified by: Payload inspection
FR-030	Functional	The system shall write an append-only audit entry recording the actor, action, affected entity, timestamp and changed values for every creation, modification and deletion of a user, question, examination, answer key or result.	Medium	Pass: editing a question creates one audit row with old and new values; no screen or endpoint can edit or delete an audit row.	Disputes need an untampered record of who changed what.	UC-13 · Sprint 2 · Verified by: Security audit

Functional requirement count: 30 — Sprint 1: 19 · Sprint 2: 11.

3. Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
NFR-001	Non-functional — Product (Performance)	The system shall complete 95% of page loads and requests within 2 seconds while 200 attempts are active concurrently.	High	Pass: a Locust load test with 200 simulated students shows a 95th-percentile response time ≤ 2 s.	A slow paper costs students time.	Sprint 2
NFR-002	Non-functional — Product (Performance)	The system shall evaluate a submitted 100-question attempt within 5 seconds.	High	Pass: a timed test of evaluating a 100-question attempt completes in ≤ 5 s.	Results must be ready right after the exam.	Sprint 2
NFR-003	Non-functional — Product (Capacity)	The system shall sustain at least 200 concurrent attempts and a question bank of at least 50,000 questions without breaching NFR-001.	Medium	Pass: NFR-001's load test passes with the bank seeded to 50,000 questions.	A department-wide exam must not degrade the system.	Sprint 2
NFR-004	Non-functional — Product (Reliability)	After an unplanned server restart, the system shall restore service within 60 seconds, with every in-progress attempt resumable with its saved responses and correct remaining time.	High	Pass: killing and restarting the server during an attempt restores service in ≤ 60 s and the attempt resumes intact.	A crash must not void an examination.	Sprint 2
NFR-005	Non-functional — Product (Portability)	The system shall work on the two latest versions of Chrome, Edge and Firefox, and remain usable at viewport widths of 360 px and above.	Medium	Pass: the attempt flow completes on each browser and at 360 px width without horizontal scrolling.	Students use their own devices.	Sprint 2
NFR-006	Non-functional — Organisational (Organisational)	All code and documents shall be held in one Git repository; work shall proceed on feature branches; every merge to `main` shall be through a pull request raised by the assigned member and reviewed by another member.	High	Pass: the GitHub history shows no direct commits to `main` after 19 Oct, and every merged PR has a reviewer other than its author.	Required by the course; keeps `main` working.	Sprint 1
NFR-007	Non-functional — Organisational (Organisational)	Every functional requirement shall be linked to at least one test case in the Test Plan.	Medium	Pass: the coverage table of `Documents/Test_Plan.pdf` lists all 30 FRs.	Untested requirements are unverified claims.	Sprint 2
NFR-008	Non-functional — External	Personal data shall be handled per the Digital	Low	Pass: registration cannot complete	Indian law applies to	Sprint 2

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
	(Legal)	Personal Data Protection Act, 2023: consent at registration, collection only for examination use, and erasure on request within 30 days.		without ticking consent; an Administrator can erase a Student's personal data.	student data.	
NFR-009	Non-functional — External (Accessibility)	The attempt flow shall meet WCAG 2.1 Level AA: fully keyboard-operable, text contrast at least 4.5:1, and every option control labelled for screen readers.	Medium	Pass: an axe/Lighthouse audit reports no AA violations, and the whole attempt can be completed with the keyboard alone.	Every student must be able to sit the exam.	Sprint 2
NFR-010	Non-functional — External (Maintainability)	Examination policy values — window, pass mark and negative marking — shall be configurable per examination and shall not be hard-coded.	Medium	Pass: code review finds no hard-coded pass mark or negative-marking value; changing them on an examination changes its scoring.	Departments use different rules.	Sprint 2
NFR-011	Non-functional — Product (Security)	Passwords shall be stored only as salted hashes using an adaptive algorithm, and shall never be stored or logged in plain text.	High	Pass: the password column holds only `pbkdf2_sha256$…` hashes, and the logs contain no password after a registration and a sign-in.	A database leak must not leak passwords.	Sprint 1
NFR-012	Non-functional — Product (Security)	The system shall lock an account for 15 minutes after 5 failed sign-in attempts within 15 minutes.	Medium	Pass: the 6th sign-in attempt after 5 failures is refused even with the right password, and succeeds after 15 minutes.	Blocks password guessing.	Sprint 2
NFR-013	Non-functional — Product (Security)	Every state-changing request shall carry an anti-CSRF token, all database access shall go through the ORM (no string-built SQL), and error pages shall show no stack trace or internal detail.	High	Pass: a POST without a CSRF token is refused with HTTP 403; a code search finds no raw SQL; a forced error shows a generic page with `DEBUG=False`.	Covers the most common web attacks.	Sprint 1
NFR-014	Non-functional — Product (Security)	Secrets such as the Django `SECRET_KEY` and database credentials shall not be committed to the repository; they shall be read from environment variables.	High	Pass: a secret scan of the public repository finds no secret, and the app refuses to start without the `SECRET_KEY` variable.	The repository is public.	Sprint 1

Non-functional requirement count: 14 — Sprint 1: 4 · Sprint 2: 10.

4. Assumptions, Dependencies and Open Items

Assumptions and dependencies. Students have a personal device with a supported browser and an internet connection · examinations are attempted remotely · the server clock is synchronised by NTP, since every timing rule depends on it · the system is a single Django application with an SQLite database, deployed on one Linux virtual machine for the demonstration (see `Documents/SAD.pdf`).

Open items awaiting the course instructor; the assumption taken is used throughout this SRS and the Test Plan:

Open question	Affects	Assumption taken
Negative marking — a flat deduction per wrong answer, or a fraction of the question's marks	FR-023, UC-11	**Flat** deduction of the question's configured negative marks
Is a score exactly at the pass mark a Pass?	FR-024, UC-11	**Inclusive** — 40.0% against a 40% pass mark is a Pass

5. Traceability

Artefact	Where
Use case diagram — actors, UC-01 … UC-13 and their relationships	`Documents/Use_Case_Diagram.pdf`
Use case flows — main, alternate and exception flows for UC-01 … UC-13	`Documents/Use_Case_Flow.pdf`
Backlog — one GitHub issue per FR / NFR, with story points, assignee and sprint	GitHub Project *Online Exam Portal - Backlog*
Sprint 1 and Sprint 2 plan	`Documents/Sprint_Plan.pdf`
Architecture — components realising each requirement group	`Documents/SAD.pdf`
Test cases for every functional requirement (118 cases, 13 use cases)	`Documents/Test_Plan.pdf`